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Interstitial Cystitis/Bladder Pain Syndrome (IC/BPS) Diagnosis: Current Limitations and a Pragmatic Clinical Diagnostic Definition

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ABSTRACT

Objective: To develop a consensus on diagnostic criteria for interstitial cystitis/bladder pain syndrome (IC/BPS).

Materials and Methods: A subcommittee was identified based on expertise in IC/BPS diagnostic criteria. An outline was generated and iteratively modified until it was found to be acceptable by subcommittee members as the basis for manuscript generation. The manuscript was presented and revised in two iterations according to feedback from international key opinion leaders at the Global Consensus on IC/BPS and the AUA Annual Meeting, respectively.

Results: The patient history and physical examination are necessary components in the diagnosis of IC/BPS. Urinalysis and urine culture are necessary laboratory tests to rule out exclusionary conditions including active infection. The National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) criteria, which were established in 1988 for research purposes, pose several limitations and result in the exclusion of a large proportion of IC/BPS patients when applied clinically. Thus, we put forth a pragmatic and streamlined definition that is aligned with existing clinical guidance and standard diagnostic workup.

Conclusions: The clinical diagnosis of IC/BPS is based on history, physical examination, and urine studies. IC/BPS is clinically defined as an unpleasant sensation (e.g. pain, discomfort, pressure, burning) that worsens with bladder filling and improves with bladder emptying, of 3 or more months duration, in the absence of exclusionary diagnoses that would likely account for the symptomatology. A substantial number of IC/BPS patients have comorbid pelvic disorders (e.g., pelvic floor dysfunction, vulvodynia, endometriosis) which require separate treatment.

Trial Registration: This study is not a clinical trial and thus does not warrant registration as such.

1 | Introduction

Interstitial cystitis/bladder pain syndrome (IC/BPS) is a chronic condition that generally includes symptoms of bladder-associated pain, pressure, and/or discomfort [1]. Historically the condition was believed to affect women about ten times more commonly than men, but more recent data support a narrower

ratio [2–4]. It is estimated that there is a prevalence of IC/BPS in females in the U.S. between 2.7% and 6.5% (based on whether a high sensitivity or high specificity is used) [3]. It is projected that between 3.3 and 7.9 million adult women in the U.S. have symptoms consistent with a possible diagnosis of IC/BPS. In men in the U.S., a prevalence of IC/BPS is estimated between 1.9% (high specificity definition) and 4.2% (high sensitivity

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definition) [2]. This translates to a prevalence between 2.1 and 4.6 million adult U.S. men. The median age of onset has often been reported in the mid 40s [5], but onset can be much earlier. The age of diagnosis is typically 30 to 70 years of age [6].

IC/BPS is commonly associated with widespread pain and comorbid conditions, and thus diagnostic criteria aim to distinguish IC/BPS from these other, often overlapping conditions. The formal classification and nomenclature of IC/BPS has evolved over time as research has expanded and patient perceptions have been more adequately delineated. There is no universally-accepted set of diagnostic criteria for IC/BPS. Here, we seek to provide an overview of the current diagnostic criteria for IC/BPS based on primary literature, historical perspectives, contemporary guidelines, and expert opinion. We describe limitations of the diagnostic criteria of the NIDDK from 1988. Finally, we provide contemporary clinical criteria, supported by expert consensus. The criteria do not possess the same limitations of the research definition that has been unmodified, and often applied clinically, for nearly 40 years.

2 | Materials and Methods

In the Global Consensus Meeting on IC/BPS 2025, nine subcommittees were developed based on expertise in respective areas, and each tasked with a particular topic related to IC/BPS. This manuscript describes the efforts of the “Diagnostics” group of 6 subcommittee. The group generated an outline, which was iteratively updated to include input from all members, until the outline in its final form was mutually agreed upon as a basis for manuscript generation. From that outline, the manuscript was written and submitted to the larger Global Consensus on IC/BPS committee for review. The manuscript was presented at the Global Consensus on IC/BPS in April 2025, and feedback from the Committee consisting of international key opinion leaders in the field, and others in attendance, was implemented. The document was revised based on feedback from the group, and the revised version was presented at the Society for Infection and Inflammation in Urology session at the 2025 American Urological Society Annual Meeting in Las Vegas, Nevada, USA. Feedback from the Committee and others in attendance at the session was again incorporated and the final version of the manuscript generated.

2.1 | Symptoms and Clinical Assessment

2.1.1 | Characteristic Symptoms

The hallmark symptom is pain arising from the urinary bladder. This is typically localized to the suprapubic region, but it can also radiate to the urethra and/or perineum. The sensation may be prescribed as any of the following: pain, pressure, or discomfort [7]. Of note, a subset of patients does not report pain. For example, epidemiologic investigations have demonstrated that some individuals with IC/BPS report severe urinary frequency/urgency, and may describe their sensation as a sensation of discomfort, pressure or burning, and not necessarily pain [7]. However, even in these patients there is clearly an “unpleasant sensation” which is causing them to feel the need

to urinate very frequently, in which case their symptoms would meet the recognized criteria for ‘pain’ even if not acknowledged by the patients.

The pain or other unpleasant sensation is generally described as associated with the voiding cycle: worse as the bladder fills, and improved/relieved by voiding. This phenomenon often distinguishes IC/BPS from other etiologies of pelvic pain. In many cases, patients may also experience pain or burning with urination. Such symptomatology is consistent with urethritis, and in some cases may also be associated with IC/BPS. In addition to the unpleasant sensation (pain, burning, pressure, and/or discomfort), patients often have additional accompanying urinary symptoms. These may include urgency due to bladder pain (as opposed to urgency due to fear of leakage, which is more typical for overactive bladder than IC/BPS), frequency, and/or nocturia [8]. About 1/3 of patients experience urgency incontinence, and some may also describe urgency even after voiding. Daytime frequency is usually elevated, and most patients with IC/BPS void more than 10 times daily [9]. A bladder diary can help quantitate functional bladder capacity as well as the degree of frequency and nocturia. In IC/BPS, a common finding is low functional capacity, and nearly identical volumes per void, unlike overactive bladder wherein voided volumes may vary considerably. Nocturia, due to reduced functional bladder capacity and due to pain interrupting sleep, is also common. Assessment of nocturia should consider the entire clinical picture including sleep-inducing medications, obstructive sleep apnea, and diuretic use, in conjunction with the bladder diary. Nocturia frequency as well as duration between nocturia episodes should be considered. Of note, symptoms may fluctuate. In the RAND IC Epidemiology (RICE) study, since the time of onset of the condition, 6.2% of IC/BPS patients reported symptoms “rarely/never,” 44% “some of the time,” 20% “about half of the time,” 18% “most of the time,” and 12% “all of the time” [10]. Symptomatic flares are a well-known phenomenon in IC/BPS.

Chronicity of symptoms in IC/BPS is also an important criterion for diagnosis. The necessary duration is variable based on available guidance. The AUA IC/BPS guidance indicates a 6 week duration is necessary [11], whereas the NIDDK criteria indicate a duration of 9 months [12]. The panel consensus is that 3 months represents a reasonable cutoff to define chronic symptoms for IC/BPS.

2.1.2 | Comorbid Disorders

Comorbid disorders are pelvic conditions that may co-occur with IC/BPS, and thus may require a separate diagnosis and treatment [13]. As such, a focused history as well as physical examination are important to identify such conditions. The presence of comorbid disorders does not exclude IC/BPS. Comorbid conditions include pelvic floor hypertonia/dysfunction, and voiding dysfunction [14], vulvodynia [15], recurrent UTI [13], endometriosis [16], and urethritis [17]. In cases wherein such conditions are suspected, appropriate workup should be initiated. For example, pelvic floor hypertonia/dysfunction may be commonly co-occurring with IC/BPS, with one study reported that 87% of women with IC/BPS had findings consistent with pelvic floor

dysfunction [18]. Pelvic floor dysfunction generally has tenderness and/or banding on palpation of the pelvic floor muscles and responds well to physical therapy. Individuals with voiding dysfunction often have difficulty initiating voiding, and their symptoms may be intermittent. Those with vulvodynia may have vulvar tenderness to palpation, and those with endometriosis, particularly when involving the bladder, may have similar symptoms that aggravate during menses. In some cases, they may have cyclical hematuria.

2.1.3 | Exclusionary Disorders

There is an important distinction between comorbid disorders and exclusionary disorders of IC/BPS. For example, individuals with IC/BPS may also suffer from recurrent UTIs or voiding dysfunction. To the contrary, exclusionary disorders are pelvic conditions that are known to cause pain and typically *exclude* the diagnosis of IC/BPS. This is especially true for research studies. Like for comorbid disorders, the focused history and physical examination are important in the identification of exclusionary disorders. As discussed subsequently, ancillary testing may be employed when exclusionary disorders are suspected. Exclusionary disorders may be categorized as either current or prior/current diagnoses. Current exclusionary disorders include active pelvic infection (e.g. acute bacterial cystitis, sexually transmitted infection, vaginitis, epididymitis, prostatitis), bladder or ureteral calculi, urethral diverticulum, urethral stricture, and neurologic causes of the pain (e.g., nerve entrapment, herniated disc). Current/prior exclusionary disorders, which may be historical or active for a given patient, include pelvic radiation, gynecological cancers, bladder cancer/carcinoma in situ, chemotherapy-induced cystitis, ketamine cystitis (recreational ketamine use is a global phenomenon and should be asked about explicitly while obtaining a history), or tuberculous cystitis.

2.1.4 | Physical Examination Findings

The physical examination (including abdominal exam and pelvic examination, with rectal examination in men) is performed with the focus of identifying comorbid disorders and exclusionary disorders that may contribute to the current symptomatology. Important portions of examination include evaluation for pain derived from the pelvic floor musculature and, in women, vulvar dermatoses and/or tenderness. Transabdominal or transvaginal palpation of the bladder may reproduce the pain/discomfort associated with IC/BPS. Pain during bladder palpation is consistent with the diagnosis of IC/BPS [8], but may not be apparent in cases wherein only a distended bladder may elicit symptoms.

2.1.5 | RAND Interstitial Cystitis Epidemiology (RICE) IC/BPS Case Definition

As part of a large population based epidemiologic study of IC/BPS, a standard case definition for IC/BPS was developed with known sensitivity and specificity values in a sample of women with IC/BPS (cases) and with conditions similar to IC/BPS (overactive bladder, vulvodynia, endometriosis [controls]) [19].

TABLE 1 | RICE Case Definition Questionnaire Items.

(Sensitivity = 81%, Specificity = 54%)

1. In the past 3 months, have you ever had a feeling of pain, pressure, or discomfort in your lower abdomen or pelvic area—that is, the part of your body that is above your legs and below your belly button?

< 1 > YES

< 2 > NO

2. In the past 3 months, have you had a feeling of a strong urge or feeling that you had to urinate or “pee” that made it difficult for you to wait to go to the bathroom?

< 1 > YES

< 2 > NO [go to Q4]

3. Would you say this urge to urinate is mainly because of pain, pressure or discomfort or mainly because you are afraid you will not make it to the toilet in time to avoid wetting?

< 1 > PAIN, PRESSURE, DISCOMFORT, OR

< 2 > FEAR OF WETTING

4. In the past 3 months, before you urinate, as your bladder starts to fill, does your feeling of pain, pressure, or discomfort usually:

< 1 > Get worse,

< 2 > Get better, or

< 3 > Stay the same?

5. In the past 3 months (when you were having symptoms), how many times on average have you had to go to the bathroom to urinate during the day when you are awake?

ENTER NUMBER OF TIMES: _____

To meet IC/BPS diagnostic criteria requires the presence of pain plus EITHER urgency or frequency

Pain = “Yes” answer to questions 1 and 4

Urgency = “Yes” answer to question 2 plus response of “pain, pressure or discomfort” for question 3

Frequency = response of 10 times or greater on question 5

For example:

- if a respondent reports pain (‘yes’ response to questions 1 and 4) PLUS frequency of 12+ on question 5, they met the criteria regardless of their answers to questions 2 and 3
- if a respondent reports pain (‘yes’ response to questions 1 and 4) PLUS urgency on question 2 that was due to “pain, pressure or discomfort” on question 3, they met the criteria regardless of their answer to question 5
- if a respondent does not answer “yes” to question 1 AND question 4, they did not meet criteria regardless of their answers to the other questions

This case definition had a sensitivity of 81% and a specificity of 54%, and the criteria are presented in Table 1. While not intended for clinical use, this case definition may be useful when developing diagnostic criteria for IC/BPS.

2.1.6 | Patient-Reported Outcome Measures/Questionnaires

Patient-reported outcome measures (PROM) and questionnaires will be covered elsewhere. Questionnaires related to IC/BPS are typically used for symptom characterization and severity, or measuring response to treatment. They may be useful adjuncts for IC/BPS, but are not, themselves, diagnostic of the condition. However it is important to administer a PROM at baseline in order allow for symptom tracking over time to assess for treatment response.

2.2 | Diagnostic Testing

2.2.1 | Urinalysis and Urine Culture

The urinalysis and urine culture are standard diagnostic tests performed during the workup of IC/BPS. A urine culture and urinalysis should both be performed [11]. The urine studies may help to identify or rule out exclusionary disorders. For example, nitrites or leukocyte esterase on urinalysis, or a uropathogen isolated on culture may be consistent with an active UTI (in the context of symptoms). Of note, UTI may result in an IC/BPS flare, but as opposed to recurrent UTI, IC/BPS symptoms typically persist between two infections. Additionally, microscopic hematuria as detected on the urinalysis should be appropriately evaluated. In cases wherein the post-void residual volume of the bladder is elevated, obstructive causes of bladder symptoms should be pursued.

2.2.2 | Diagnostic Cystoscopy

The diagnostic cystoscopy is an important diagnostic test that should be used early in the diagnostic process for IC/BPS. The utility of the cystoscopy lies in its ability to identify the presence of Hunner lesions [20], which are present in about 10% of IC/BPS cases [21]. Hunner lesions are distinctive, inflamed appearing erythematous patches often with small vessels that radiate toward a central ulceration or scar. It is common for a biopsy of these lesions to be performed to exclude malignancy, although they typically appear distinct from malignant lesions. Any suspicious lesion may be biopsied to rule out malignancy. though biopsy of a Hunner's lesion is not recommended as it is not diagnostic. The identification of a Hunner lesion on cystoscopy essentially confirms a diagnosis of IC/BPS, and is important for therapeutic planning since the Hunner lesion subtype of IC/BPS may have different clinical characteristics [22] and responds to unique treatment modalities [11]. Cystoscopy may also identify other exclusionary disorders including carcinoma-in-situ (CIS). Cystoscopy may reproduce a patient's discomfort with bladder filling (when done under local anesthesia), and the tip of the scope may be used to determine the regions of discomfort. Other than cystoscopy for Hunner lesions, there is no single test that reliably identifies the presence or absence of IC/BPS.

2.2.3 | Cystoscopy with Hydrodistension

Cystoscopy with hydrodistension, generally employed as a therapeutic modality, is rarely performed solely for diagnostic purposes given the lack of additional clinical information it

provides beyond that obtained through history and physical examination [23]. The presence of glomerulations (pinpoint petechial hemorrhages) following hydrodistension had historically been a requirement to diagnose IC/BPS, but it has since been demonstrated that many patients with IC/BPS do not have glomerulations [24]. Additionally, even patients without IC/BPS may have glomerulations following repeated bladder distention [25]. Though glomerulations are not sufficient for diagnosis, their presence may support a diagnosis of IC/BPS, particularly in the context of other findings such as a small capacity bladder that bleeds after hydrodistension. Additionally, cystoscopy may also assess anesthetic bladder capacity, which has been shown to distinguish between non-bladder-centric (high capacity) and bladder-centric (low capacity) IC/BPS phenotypes [26].

2.2.4 | Potassium Sensitivity Testing and Anesthetic Challenge

The potassium sensitivity test and anesthetic challenge, while both imperfect, are designed to identify the *hyperesthetic* bladder. The potassium sensitivity test (PST) [27] is based on the theory that IC/BPS is caused by a dysfunctional bladder surface mucosal layer which allows urinary solutes (such as potassium) to migrate into the bladder interstitium and cause pain [28]. The PST measures the difference in patient response to instillation of a water solution versus a potassium solution. Patients who have an increase in response following the potassium, but not water, solution, are positive. Of note, the PST is positive in other disorders including radiation cystitis [29], acute bacterial cystitis [30], prostatitis [31], and overactive bladder [32]. In addition, in one study 24% of individuals who met the NIDDK criteria for IC/BPS had a negative PST [33]. Many clinicians prefer not to perform the PST because of this lack of specificity for IC/BPS and its tendency to induce pain.

An alternative approach to the PST is an anesthetic challenge where a lidocaine-based solution is instilled into the bladder. In this instance, a reduction in pain is considered characteristic of IC/BPS. Since an anesthetic challenge does not cause pain, and since lidocaine instillations may be therapeutic as well as diagnostic [34–36], this approach is preferred by many clinicians.

It is important to acknowledge that the response to the PST or an anesthetic challenge is not by itself diagnostic of IC/BPS, and that neither test has been shown to be predictive of efficacy for any specific therapy. As a result, the clinical utility of these tests is limited at this time.

2.2.5 | Urodynamic Testing

Urodynamic testing is not a mandatory test for the diagnosis of IC/BPS but may be performed in certain patients where bladder outlet obstruction, voiding dysfunction, or other comorbid or exclusionary disorders are suspected. In patients with IC/BPS, urodynamic testing often demonstrates pain with bladder filling (hypersensitivity) and reduced bladder capacity [37]. Other findings, such as detrusor overactivity, stress urinary incontinence, or bladder outlet obstruction do not exclude IC/BPS, as these conditions may be comorbid with IC/BPS and require separate

diagnosis and treatment, as described above. Urodynamics testing results have not been shown to correlate with IC/BPS treatment response.

3 | National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) Diagnostic Criteria and Associated Limitations

In 1988, the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) established consensus research criteria [12]. These criteria were initially designed as criteria for research studies, to be used to ensure homogenous cohorts across investigations. However, over time they were adopted, erroneously, as

clinical diagnostic criteria. When applied clinically, the criteria exclude a large proportion of patients with IC/BPS. The numerous limitations associated with the NIDDK criteria are summarized in Table 2.

3.1 | Contemporary Clinical Criteria for the Diagnosis of IC/BPS

Given the limitations of the NIDDK criteria for the clinical diagnosis of IC/BPS, together with new understanding of the pathophysiology of the disease and patient experiences, we put forth a set of criteria for the diagnosis of IC/BPS. The criteria are pragmatic, aligned with the available guidance from the

TABLE 2 | 1988 National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) criteria and associated limitations for interstitial cystitis/bladder pain syndrome (IC/BPS) diagnosis.

NIDDK criterion	Limitations
Automatic inclusion	
Hunner lesions on cystoscopy	Cystoscopy not always indicated in diagnostic workup
Positive factors	
Pain with bladder filling and relief with emptying	Not all IC/BPS patient describe pain. Some describe the unpleasant sensation as pressure, discomfort, and/or burning
Suprapubic, pelvic, urethral, vaginal, or perineal pain	See above
≥ 10 glomerulations on each of 3 of 4 bladder quadrants endoscopy	Glomerulations are now known to be a nonspecific finding (absent in up to half of individuals with IC/BPS)
Reduced compliance on cystometrogram	Exceedingly rare in the absence of neurologic disease or prior pelvic radiation. Requires expensive urodynamics testing.
Exclusionary criteria	
< 18 years of age	Reports of IC/BPS in children and adolescents
Benign or malignant bladder tumors	
Radiation cystitis	
Tuberculous cystitis	
Bacterial cystitis	Can co-occur with IC/BPS Often poorly documented (no urine cultures, etc.)
Vaginitis	
Cyclophosphamide cystitis and other chemical cystitis	
Symptomatic urethral diverticulum	
Uterine, cervical, vaginal, urethral cancer	
Active genital herpes	
Bladder or lower ureteral calculi	
Daytime frequency < 5 times in 12 h	
Nocturia < 2 time nightly	
Symptoms relieved by antibiotics, urinary antiseptics, urinary analgesics (e.g. pyridium)	Difficult to operationalize Potential/likely diffuse mechanism of action of these medications
Duration < 9 months	Arbitrary; does not align with AUA guidance
Involuntary contractions on urodynamics	Detrusor overactivity is now considered to be a comorbid condition rather than an exclusionary condition. In other words, DO may coincide with IC/BPS
Capacity > 400 cc, absence of sensory urgency	Arbitrary cutoff, imprecise definition of sensory urgency. Bladder capacity data can be obtained noninvasively via voiding diary.

TABLE 3 | Contemporary criteria for clinical diagnosis of IC/BPS^a.

Criterion
An unpleasant sensation (e.g. pain, discomfort, pressure, burning) that worsens with bladder filling and/or improves with bladder emptying
At least one of the following symptoms:
– Urgency due to bladder discomfort (rather than fear of leakage)
– Abnormal daytime urinary frequency (10+)
Symptom duration of 3 or more months
The absence of exclusionary disorders that could account for the symptomatology

^aInitial workup for suspected IC/BPS should include history, physical examination, urinalysis and urine culture. Additional ancillary testing may be performed on a patient-by-patient basis to rule out suspected comorbid or exclusionary conditions. IC/BPS, interstitial cystitis/bladder pain syndrome.

AUA [11], and designed to be identified using only the essential diagnostic workup of history, physical examination, and urinalysis with culture, but no other ancillary testing [38, 39]. Ancillary testing may be pursued on a case-by-case basis in situations wherein Hunner lesions, or comorbid or exclusionary disorders are suspected. The contemporary clinical criteria for IC/BPS diagnosis are listed in Table 3.

4 | Conclusion

We provide an overview of the diagnostic criteria for IC/BPS. We discuss the history, physical examination, necessary testing and ancillary testing. We describe the NIDDK criteria and their associated limitations for diagnosis of IC/BPS. We provide a contemporary set of IC/BPS diagnostic criteria, which aligns with the available guidance and diagnostic workup of the condition. The clinical diagnosis of IC/BPS is based on history, physical examination, and urine studies. *IC/BPS is clinically defined as an unpleasant sensation (e.g. pain, discomfort, pressure, burning) of at least 3 months duration, that worsens with bladder filling and/or improves with bladder emptying, with concomitant pain-driven urinary urgency or abnormal daytime frequency, in the absence of exclusionary diagnoses that would likely account for the symptomatology.*

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The authors have nothing to report.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

Glenn T. Werneburg: consultant: Light Line Medical, Atterx Biotherapeutics, Clarametix Biosciences; Medical Advisory Board: Renascent Diagnostics. Robert Moldwin: consultant: Glycologix. J. Quentin Clemens: Royalties - UpToDate, Stock ownership - Merck, Consultant - Abbvie, Medtronic, Boston Scientific, Glaxo Smith Kline, Iota-Astellas, ProvePharm. The other authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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